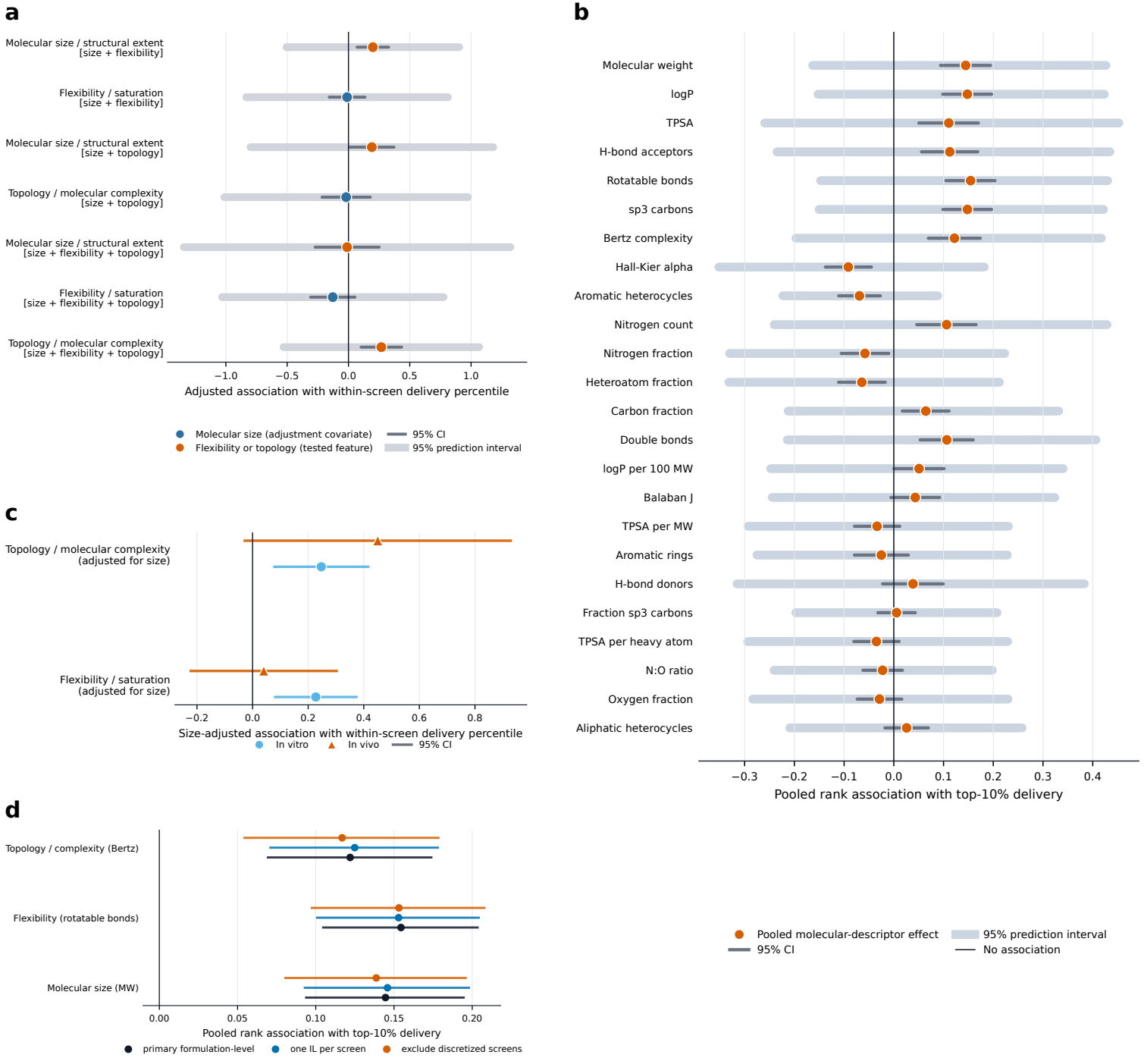

a
Molecular size / structural extent [size + flexibility]
Flexibility / saturation [size + flexibility]
Molecular size / structural extent [size + topology]
Topology / molecular complexity [size + topology]
Molecular size / structural extent [size + flexibility + topology]
Flexibility / saturation [size + flexibility + topology]
Topology / molecular complexity [size + flexibility + topology]
−1.0 −0.5 0.0 0.5 1.0
Adjusted association with within-screen delivery percentile
Molecular size (adjustment covariate)
Flexibility or topology (tested feature)
95% CI
95% prediction interval
b
Molecular weight
logP
TPSA
H-bond acceptors
Rotatable bonds
sp3 carbons
Bertz complexity
Hall-Kier alpha
Aromatic heterocycles
Nitrogen count
Nitrogen fraction
Heteroatom fraction
Carbon fraction
Double bonds
logP per 100 MW
Balaban J
TPSA per MW
Aromatic rings
H-bond donors
Fraction sp3 carbons
TPSA per heavy atom
N:O ratio
Oxygen fraction
Aliphatic heterocycles
−0.3 −0.2 −0.1 0.0 0.1 0.2 0.3 0.4
Pooled rank association with top-10% delivery
Pooled molecular-descriptor effect
95% CI
95% prediction interval
No association
c
Topology / molecular complexity (adjusted for size)
Flexibility / saturation (adjusted for size)
−0.2 0.0 0.2 0.4 0.6 0.8
Size-adjusted association with within-screen delivery percentile
In vitro
In vivo
95% CI
d
Topology / complexity (Bertz)
Flexibility (rotatable bonds)
Molecular size (MW)
0.00 0.05 0.10 0.15 0.20
Pooled rank association with top-10% delivery
primary formulation-level
one IL per screen
exclude discretized screens